

The `stroke()` function in ProcessingJS sets the color used to draw lines and borders around shapes. It remains active for all subsequent drawing commands until you change it or disable it.

Syntax Variations

- `stroke(rgb)` – Grayscale value (0 to 255).
- `stroke(rgb, alpha)` – Grayscale with transparency.
- `stroke(r, g, b)` – Red, Green, and Blue values.
- `stroke(r, g, b, alpha)` – RGB with transparency.
- `stroke(hex)` – Hexadecimal color code (e.g., `#FF0000`).

```
function draw() {  
  background(255); // white background  
  
  strokeWeight(4); // Thicken the lines  
  
  stroke(255, 0, 0); // Red stroke  
  rect(20, 20, 50, 50);  
  
  stroke(0, 0, 255, 100); // Semi-transparent blue stroke  
  line(0, 0, 100, 100);  
}
```

Key Related Functions

- `noStroke()` – Disables the border entirely.
- `strokeWeight(pixels)` – Changes the thickness of the line.
- `fill(r, g, b)` – Sets the inner color of the shape.

https://processing.org/reference/stroke_.html

https://gotoloop.github.io/processing-js.github.io/reference/stroke_/



The `arc()` function in Processing draws a section of an ellipse. It requires the (x, y) position, width, height, start and end angles (in radians), and an optional rendering mode. Angles are measured clockwise, with 0 pointing right.

Syntax

```
arc(x, y, width, height, start, stop, [mode]);
```

Parameters

- **x, y:** The x and y coordinates of the arc's center (or the top-left corner, depending on `ellipseMode`).
- **width, height:** The width and height of the bounding box of the ellipse.
- **start, stop:** The angles to start and stop drawing the arc, measured in radians (always drawn clockwise).
- **mode (*optional*):** Defines how the arc is closed.
 - **OPEN:** Default mode; the arc is left open.
 - **PIE:** Closes the arc by drawing lines from the center to the start and end points (like a slice of pizza).
 - **CHORD:** Closes the arc by drawing a straight line directly from the start point to the end point.

java

```
void setup() {  
  size(600, 200);  
  background(255);  
  
  // OPEN arc  
  fill(150);  
  arc(100, 100, 100, 100, 0, PI, OPEN);  
  
  // PIE arc  
  fill(200);  
  arc(250, 100, 100, 100, 0, PI, PIE);  
  
  // CHORD arc  
  fill(250);  
  arc(400, 100, 100, 100, 0, PI, CHORD);  
}
```

<https://processing.org/tutorials/curves>

https://processing.org/reference/arc_.html



```
var Rainbow = function(x,y,radius) {  
  this.x=x;  
  this.y=y;  
  this.radius=radius;  
};
```

```
var drawRainbow = function(rainbow) {  
  noFill();  
  strokeWeight(5);  
  stroke(255, 0, 0);  
  arc(rainbow.x, rainbow.y, rainbow.radius, rainbow.radius, 180, 360);  
  stroke(250, 200, 0);  
  arc(rainbow.x, rainbow.y, rainbow.radius-10, rainbow.radius-10, 180, 360);  
  stroke(240, 255, 102);  
  arc(rainbow.x, rainbow.y, rainbow.radius-20, rainbow.radius-20, 180, 360);  
  stroke(29, 255, 13);  
  arc(rainbow.x, rainbow.y, rainbow.radius-30, rainbow.radius-30, 180, 360);  
  stroke(122, 155, 255);  
  arc(rainbow.x, rainbow.y, rainbow.radius-40, rainbow.radius-40, 180, 360);  
};
```

```
var Rainbow1 = new Rainbow(150,200,250);  
drawRainbow(Rainbow1);
```

```
var Rainbow2 = new Rainbow(250,300,300);  
drawRainbow(Rainbow2);
```